

Learning and Assessment

This session explored how AI is reshaping learning and assessment in STEM. Faculty discussed how to maintain rigor and cognitive development while leveraging AI for personalization, feedback, and engagement.

Key Findings from Faculty Discussion

- **Employability & Preparedness:** Recent graduates with AI literacy report stronger job alignment and faster employment, but concerns remain over how long AI skill advantages will last before the lack of core disciplinary mastery becomes limiting.
- **Pedagogical Experimentation:** Assignments comparing student-generated versus AI-generated work reveal that students cannot meaningfully critique AI outputs without mastering underlying methods.
- **Cognitive Erosion:** Faculty noted visible differences in analytical depth when students rely heavily on AI for problem solving; raising the question of whether over-reliance bypasses mastery in STEM.
- **Assessment Redesign:** Ideas included incorporating reflection prompts (“What did you learn to do better?”), student-generated challenge questions, and low-stakes quizzes to measure process and growth.
- **Academic Integrity Reframed:** A few faculty warned that there is an epidemic of student cheating with AI at WPI without meaningful actions by the institution. Others argued that the concept of cheating is relative to the assignment and whether traditional notions of 'cheating' remain relevant.

References

[Design and assessment of AI-based learning tools in higher education \(2025\) \(link\)](#)

Structured and intentional AI integration can enhance personalization and feedback, offering scalable learning support for students while maintaining instructor oversight.

[Evaluating AI-Powered Learning Assistants in Engineering Higher Education \(2025\) \(link\)](#)

AI assistants improved engagement and comprehension among engineering students, with many reporting higher efficiency and deeper understanding of technical concepts.

[Artificial intelligence skills and their impact on the future workforce \(2025, PMC\) \(link\)](#)

Graduates with advanced AI literacy show stronger alignment with their fields and greater employability, highlighting the importance of AI competencies in higher education.

[Talk is cheap: why structural assessment changes are needed for a time of GenAI \(2025\) \(link\)](#)

Generative AI challenges assessment validity, enabling surface-level completion of tasks without genuine understanding — prompting a call for reimagined assessment practices.

[The effects of over-reliance on AI dialogue systems on students \(2024\) \(link\)](#)

Excessive dependence on AI chat systems can weaken students’ cognitive and analytical skills, reducing self-reliance and critical engagement with material.

[Scaffold or Crutch? Examining College Students’ Use and Views of Generative AI Tools for STEM Education \(2024\) \(link\)](#)

Many students rely on AI-generated solutions rather than reasoning through problems themselves, risking erosion of fundamental STEM problem-solving abilities.

LEARNING and ASSESSMENT

graduates w/ AI skills are more likely to be hired → how long will this last?

graduates demand widespread adoption of AI.

Assignments where students do something (create code)
and compare to A.I. solution

Then - assignment where students only use A.I.
and try to assess what A.I. did

Notice the difference (i.o.w.: you can't critique A.I.

catch effect?

without knowing how to do it)

erosion of core cognitive skills? (Gerlich, 2025) Societies

~~bypassing~~ bypassing mastery in STEM?

how do you redesign assessment to capture process rather than answer?

Reflection → What did you learn to do? What have you learned to do better?

Challenge questions created by students

Hands-on activities

How are you growing?

Essential Moderation Knowledge: AI, Mastery, and Academic Integrity

I. The Promise of Intentional AI Integration and Enhanced Learning

- **AI as a Pedagogical Ally:** Structured and intentional AI integration offers benefits such as providing **personalized learning experiences** and **real-time feedback**. These tools serve functions like assessment/evaluation, personalized guidance, recommendations, and intelligent tutoring.
- **Boosting Engagement and Efficiency:** AI learning assistants were appreciated by engineering students for their **convenience and comfort**. Nearly half of students in one study reported **greater ease using the AI tool** compared to consulting human instructors or teaching assistants.
- **Deepening Technical Comprehension:** AI tools were found to be highly helpful for **completing homework and understanding complex course concepts** in engineering higher education. Frequent use of AI tools also correlates strongly with the perception of **improved work productivity**.
- **Supporting Self-Regulation:** AI-powered tools incorporating features like adaptive feedback and counterfactual explanations have demonstrated success in boosting students' **self-regulation abilities**, including goal-setting, planning, progress monitoring, and self-evaluation.

II. Student Preparedness and Employability Alignment

- **AI Literacy as a Differentiating Factor:** Graduates who possess greater **knowledge and more frequent use of AI tools** (especially generative ones like ChatGPT) are statistically more likely to be employed in sectors related to their major and perceive **better professional alignment**.
- **Urgent Need for Transversal Skills:** There is a nearly unanimous demand from graduates (87.2% affirmative) that universities should strengthen AI teaching across *all* majors, viewing it as a **transversal skill** rather than specialized content.
- **Widespread Feeling of Insufficient Preparation:** Despite recognizing AI's importance, a majority of graduates report feeling **insufficiently prepared** for labor market changes due to AI, with 52.0% feeling "somewhat prepared" and 26.7% feeling "little prepared".
- **Lifelong Learning Imperative:** The vast majority of graduates (95.6%) express strong interest in continuing professional development through courses on AI tools.

III. Critical Risks: AI Offloading Thinking and Cognitive Erosion

- **The Crutch Effect:** AI's capacity to streamline tasks leads to a risk of **over-reliance**, where users accept AI-generated outputs uncritically due to a tendency to favor efficient **cognitive shortcuts (heuristics)** over slower, analytical methods.
- **Erosion of Core Cognitive Skills:** Excessive dependence on AI dialogue systems can significantly diminish students' **critical thinking, analytical thinking, and decision-making abilities**. This dependency fosters complacency, reducing independent critical information analysis.
- **Bypassing Mastery in STEM:** In STEM education, problem-solving is central, yet **over half of surveyed college students** reported **simply inputting problems for AI to generate solutions**. This practice potentially bypasses the student's own essential problem-solving processes, risking the erosion of fundamental abilities.
- **The Transparency Challenge:** Many AI tools operate as "black boxes," obscuring their decision-making processes, which impedes critical reflection and fosters the passive, uncritical consumption of AI-generated content. Users with lower subject matter expertise are particularly prone to trusting the AI's advice, even when it is factually incorrect (hallucination).

IV. Academic Integrity: Shifting from Rules to Structural Design

- **The Validity Crisis:** Generative AI fundamentally challenges **assessment validity** by enabling students to complete tasks without demonstrating the relevant knowledge or genuine capability. Assessments cease to serve their primary educational purpose if assessors cannot verify student competency.
- **The Failure of Discursive Changes:** Many common institutional responses (e.g., traffic light systems, declarations) rely on **discursive changes**—modifications based solely on communicating rules and guidelines about AI use. These changes are vulnerable because they depend entirely on **voluntary student compliance**.
- **The Enforcement Illusion:** Relying on discursive changes creates an "**enforcement illusion**," borrowing the language of structural, enforceable systems (like vehicular traffic lights) without possessing the institutional or technological ability to enforce compliance.
- **Detection is Unreliable:** Current AI detection technologies are unreliable, frequently suffering from **false positives and false negatives**. Plagiarism detection software was found to identify plagiarism in fewer than 15% of AI-generated essays based on existing literature.
- **Designing for Mastery through Structure:** To maintain integrity, the moderator must advocate for **structural assessment redesign**. Structural changes alter the **nature, format, or mechanics** of the task itself, making them independent of voluntary compliance. This shift should focus on reorienting assessment from evaluating only the final **output to capturing the student's process**, skill development, and attainment of understanding over time.